COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-005: Structured Cabling

Document ID	TS-INF-005
Title	Structured Cabling
Executive Sponsor	Network Services, Sr. Manager of Network Operations
Issuing Authority	Office of Information Technology (OIT)
Applies to	Any facility, site, or physical location maintained by, or in service of, an agency of the Executive Branch of the Colorado State Government.
Tier	Standard
Version	4.0
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1. Purpose

OIT is statutorily obligated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy stakeholder needs surrounding Structured Cabling, this document is submitted as the standard for technology, specifications, methodologies, and practices to provide the service.

The ultimate purpose of this standard is to describe the minimum requirements for installing structured cabling at new and existing building sites that support Local Area Networks (LAN) and Voice connectivity, so as to: accommodate the functional requirements of present and future information services; support a multi-product and multi-vendor environment; facilitate the planning and installation of cabling systems; and ensure uniformity of structured wiring and hardware infrastructure installations in all State Agency facilities.

2. Scope

This standard governs the technology, practices, and materials used when installing cabling at any facility, site, or physical location maintained by, or in service of, an agency of the Executive Branch of the Colorado State Government.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews and approves the Standard (this document).

Network Services - Sr. Manager of Network Operations

- Owner of Standard; engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

Network Operations and Engineers

- Review floor plans and possible site surveys.
- Initiate cabling quotes with approved cabling vendors.
- Verify cabling is completed, review test reports, and update port mapping floor plans.
- Set up network configuration after completing cabling.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Entrance Facilities (EF)

- Must conform to ANSI/TIA/EIA-569 requirements.
- At least two (2) four-inch underground conduits from the existing communication pedestal to the DMARC.
- Conduit shall be installed using sweeps, not elbows, with no more than three bends and total bends not exceeding 270 degrees.
- The conduit shall provide a pre-installed pull-string.
- A minimum 4x4 plywood backboard is required in DMARC areas, installed in a secure area accessible only to those authorized by OIT Network Services.

Telecommunications Rooms (TR) / Data Closet

- Consult ANSI/TIA/EIA 569 for design and provisioning requirements.
- Room penetrations must be properly firestopped in compliance with applicable codes.
- Ventilation and air conditioning (HVAC) must be present and operational 24 hours a day, 365 days a year.
- All equipment racks, cabinets, and systems must be properly connected to the independent ground bus per ANSI/TIA/EIA 607.

Horizontal Cabling

- All components of the horizontal cabling system must be from the same manufacturer.
- No horizontal cable should be exposed within the building area.
- All copper cables shall be positioned at a minimum of 4 inches from any EMI-emitting device.

- Any new cabling runs must meet the CAT6e minimum rating or better.
- All horizontal drops must be patched into a switch port so all ports are active.

Cable Length

- Maximum distance between the telecommunications outlet and the horizontal cross-connect: 90 meters.
- Maximum length of all patch cords and jumpers in the telecommunications closet: 5 meters.
- Total channel link length shall not exceed 100 meters (328 feet).

Backbone Cabling

- Hierarchical star topology — each horizontal cross-connect connects to the main or intermediate cross-connect.
- No more than two hierarchical levels of cross-connects in the backbone.
- Recognized cables: 100 ohm twisted pair; 50/125 or 62.5/125 micron multimode fiber; single-mode fiber.

Test and Certify Cabling

- Contracted vendor must test all wires with a Level III cable tester to certify conformance to ANSI/TIA/EIA-568-B specifications.
- Printed and electronic Pass/Fail results must include: electrical length, cross-talk, end-to-end attenuation, continuity, and date of test.
- Optical link testing must follow ANSI/TIA/EIA-568-B.3 for multi-mode fiber and ANSI/TIA/EIA-526-7 method A for single-mode fiber.

Port Mapping

- All ports shall be labeled at the wall port, matching the label on the patch panel in the telecommunications room.
- The agency, landlord, or general contractor will provide a digital copy of the current building floor plan.
- After installation, the cabling vendor will update the floor plan with labeled port numbers at each cable drop.

5. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of, and approved by, the "Owner of Standard" listed under Roles and Responsibilities.

Requests may be acquired by emailing the Senior Network Operations Manager and including other documentation required in standard procedures when executing a technology procurement or service deployment.

6. References

- TIA-568-C.0 through C.3, Commercial Building Telecommunications Cabling Standards
- TIA-569-B, Commercial Building Standards for Telecommunications Pathways and Spaces
- ANSI/TIA-606, Administration Standard for Commercial Telecommunications Infrastructures
- ANSI J-STD-607-A, Commercial Building Grounding and Bonding Requirements for Telecommunications

- TIA-942, Telecommunications Infrastructure Standard for Data Centers

7. Definitions

DMARC: Demarcation point — the point at which the public switched telecommunications network ends and connects with the customer's on-premises wiring.

Entrance Facilities: The entrance point for outside plant cable from telephone companies, private network cables, and other access providers.

Horizontal Cabling: The portion of the cabling system extending from the work area outlet through the wall/ceiling/floor to the patch panel in the telecommunications room.

Backbone Cabling: The interbuilding and intrabuilding cable connections between entrance facilities, equipment rooms, and telecommunications closets.

Structured Cable: Building or campus telecommunications cabling infrastructure comprising standardized smaller elements (subsystems), governed by standards specifying wiring for data centers, offices, and apartment buildings.

Telecommunications Rooms: The critical point between the work area and the Equipment Room or main cross-connect through backbone cabling and horizontal distribution.